

PROJECT REPORT

Customer Shopping Behaviour Analysis

End-to-end data analytics project using Python, SQL and Power BI

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AT A GLANCE

3,900

Customers analysed

\$233K

Total revenue

\$59.76

Average purchase

3.75

Average review rating

Business question: How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

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1. Project Overview

A leading retail company wants to better understand how its customers shop in order to improve sales, customer satisfaction, and long-term loyalty. Management has noticed changes in purchasing patterns across demographics and product categories, and wants to know which factors, such as discounts, reviews, seasons, or payment preferences, drive purchase decisions and repeat purchases.

This project answers that question end to end: the raw data is cleaned in Python, analysed with SQL, visualised in Power BI, and the results are summarised in this report and a stakeholder presentation.

Project workflow

Stage	Tool	What was produced
1. Data preparation	Python (pandas)	Cleaned dataset loaded into a PostgreSQL table
2. Data analysis	SQL (PostgreSQL)	10 queries answering key business questions
3. Visualisation	Power BI	Interactive customer behaviour dashboard
4. Communication	Word, PowerPoint	This report and a stakeholder presentation

2. Dataset Summary

The dataset `customer_shopping_behavior.csv` contains one row per customer, describing their latest purchase along with demographic and loyalty information.

Attribute	Value
Rows	3,900 (one per customer; every Customer ID is unique)
Columns	18
Data quality	37 missing values (all in Review Rating); no duplicate rows
Purchase amount	\$20 to \$100 (average \$59.76)
Customer age	18 to 70 (average 44)
Coverage	25 products in 4 categories, customers in all 50 US states

Column groups

Group	Columns
Customer demographics	Customer ID, Age, Gender, Location
Purchase details	Item Purchased, Category, Purchase Amount (USD), Size, Color, Season
Shopping behaviour	Review Rating, Shipping Type, Discount Applied, Promo Code Used, Payment Method
Loyalty indicators	Subscription Status, Previous Purchases, Frequency of Purchases

3. Data Preparation (Python)

The raw data was cleaned and transformed in the Jupyter notebook `Customer_Shopping_Behaviour_Analysis.ipynb` using pandas.

Step	What was done	Why
1. Load and explore	Loaded the CSV and reviewed the structure with <code>df.info()</code> and summary statistics with <code>df.describe()</code>	Understand data types, value ranges, and data quality
2. Handle missing values	Filled the 37 missing review ratings with the median rating of the same product category	Keeps all 3,900 rows; category medians are more accurate and not skewed by extreme ratings
3. Standardise column names	Converted names to snake_case and renamed Purchase Amount (USD) to <code>purchase_amount</code>	Easier to use in Python and SQL without quotes
4. Add age groups	Created <code>age_group</code> by splitting ages into 4 equal-sized groups (quartiles)	Compare behaviour across life stages
5. Add purchase frequency in days	Created <code>purchase_frequency_days</code> by converting frequency labels into approximate days between purchases	Turns text into a number usable in calculations
6. Remove redundant column	Confirmed <code>discount_applied</code> and <code>promo_code_used</code> match in all rows, then dropped <code>promo_code_used</code>	Removes duplicate information
7. Load into PostgreSQL	Wrote the cleaned DataFrame to the <code>customer</code> table using SQLAlchemy	Makes the data available for SQL and Power BI

Result: A clean table of 3,900 rows and 19 columns with no missing values, stored in the `customer` table of the `customer_behaviour` database.

New columns

`age_group` (quartiles of age)

Age group	Age range	Customers
Young Adult	18–31	1,028
Adult	32–44	942
Middle-aged	45–57	986
Senior	58–70	944

`purchase_frequency_days` (days between purchases)

Frequency label	Days
Weekly	7
Fortnightly, Bi-Weekly	14
Monthly	30
Quarterly, Every 3 Months	90
Annually	365

Example: filling missing ratings

```
# Fill each missing rating with the median rating of the same category
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

4. Data Analysis (SQL)

Ten business questions were answered in PostgreSQL using `customer_behaviour_sql_queries.sql`. Each result below matches the query output, followed by what it means for the business.

Q1. Total revenue by gender

Gender	Revenue
Male	\$157,890
Female	\$75,191

Finding: Male customers generate 67.7% of revenue, which mirrors their 68% share of customers (2,652 of 3,900). Average spend per purchase is almost the same (men \$59.54, women \$60.25), so the gap comes from the number of customers, not how much each one spends.

Q2. Customers who used a discount and still spent more than average

Measure	Value
Customers who used a discount	1,677
...who also spent above \$59.76	839
Share of discount users	50.0%

The query lists all 839 customers; this table summarises it.

Finding: Half of all discount users still spend above average, so discounts are not only attracting low-value purchases. These 839 customers are strong candidates for premium or exclusive offers.

Q3. Top 5 products by average review rating

Rank	Product	Average rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Finding: Gloves, Sandals and Boots are the best rated, but ratings are tightly grouped. Every product averages between 3.62 (Shirt) and 3.86, close to the overall average of 3.75, so no product clearly delights customers.

Q4. Average purchase amount: Standard vs Express shipping

Shipping type	Average purchase
Express	\$60.48
Standard	\$58.46

Finding: Express shipping customers spend \$2.02 (3.5%) more per purchase than Standard customers. The difference is small: across all six shipping types, the average ranges only from \$58.46 to \$60.73.

Q5. Subscribers vs non-subscribers

Subscribed	Customers	Avg spend	Revenue
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

Finding: Subscribers make up 27% of customers and 26.9% of revenue, and spend slightly less per purchase than non-subscribers. The subscription is not currently increasing spend per order.

Q6. Top 5 products by share of discounted purchases

Product	Discount rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Finding: About half of the purchases of these products use a discount, compared with 43.0% across all purchases. They are the most discount-dependent products and the first to review for margin impact.

Q7. Customer segments by previous purchases

Segment	Past purchases	Customers	Share
Loyal	More than 10	3,116	79.9%
Returning	2 to 10	701	18.0%
New	1	83	2.1%

Finding: Four out of five customers are Loyal (customers average 25 previous purchases). The customer base is mature, so keeping loyal customers and increasing their value matters more than acquisition alone.

Q8. Top 3 most purchased products in each category

Category	Rank	Product	Orders
Accessories	1	Jewelry	171
Accessories	2	Belt	161
Accessories	3	Sunglasses	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

Finding: Jewelry, Blouse and Pants lead with 171 orders each. Order counts are close within each category, so no single product dominates. Products with equal orders are ranked alphabetically, and Outerwear has only two products.

Q9. Subscription status of repeat buyers (more than 5 previous purchases)

Subscribed	Repeat buyers	Share
No	2,518	72.44%
Yes	958	27.56%

Finding: 27.6% of repeat buyers subscribe, compared with 22.4% of customers with 5 or fewer purchases (95 of 424). Repeat buyers are slightly more likely to subscribe, but 2,518 of them still have not, a large audience for subscription campaigns.

Q10. Revenue by age group

Age group	Total revenue	Share
Young Adult	\$62,143	26.7%
Middle-aged	\$59,197	25.4%
Adult	\$55,978	24.0%
Senior	\$55,763	23.9%

Finding: Young Adults (18–31) contribute the most revenue, but the gap between the highest and lowest group is only about \$6,400. Revenue is spread evenly across ages, so no age group should be ignored.

5. Dashboard (Power BI)

The interactive dashboard `customer_behaviour_dashboard.pbix` connects directly to the `customer` table in PostgreSQL, so it always reflects the cleaned data.

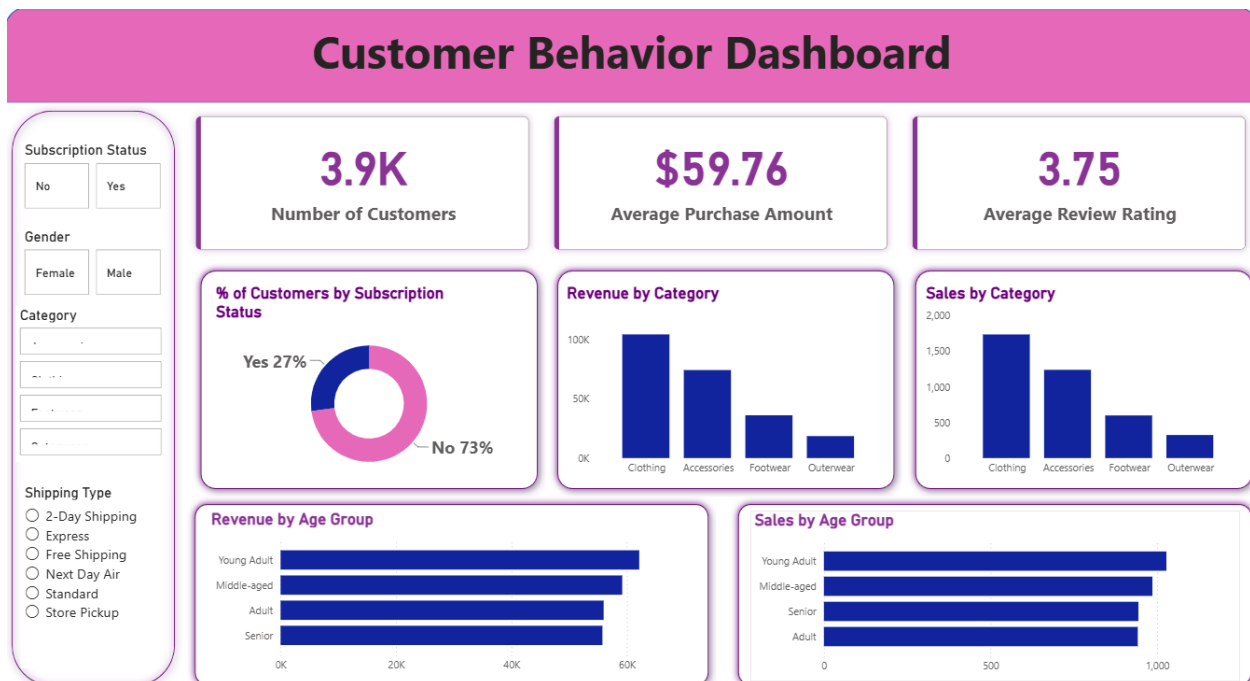


Figure 1. Customer Behaviour dashboard in Power BI

Element	Contents
KPI cards	Number of customers (3.9K), average purchase (\$59.76), average rating (3.75)
Charts	Subscription split; revenue and sales (orders) by category and by age group
Slicers	Subscription status, gender, category, shipping type
DAX measures	Number of Customers = <code>COUNT(customer_id)</code> Average Purchase Amount = <code>AVERAGE(purchase_amount)</code> Average Review Rating = <code>AVERAGE(review_rating)</code>

Revenue by category

Category	Orders	Revenue	Share
Clothing	1,737	\$104,264	44.7%
Accessories	1,240	\$74,200	31.8%
Footwear	599	\$36,093	15.5%
Outerwear	324	\$18,524	7.9%

Finding: Clothing and Accessories together generate 76.5% of revenue. Average spend per order is similar in every category (\$57 to \$60), so category revenue is driven by order volume.

6. Key Findings

- Revenue follows customer volume, not spend per order.** The average purchase stays between about \$57 and \$61 across gender, category, age group, shipping type and subscription status.
- Clothing and Accessories drive the business,** generating 76.5% of total revenue (\$233,081).
- Most customers are already loyal.** 80% have more than 10 previous purchases, and the average customer has 25.
- Subscriptions are not lifting spend.** Subscribers spend slightly less per order (\$59.49 vs \$59.87), and 72% of repeat buyers have not subscribed.
- Discounts are common but not only used by low spenders.** 43% of purchases use a discount, and half of those customers still spend above average.
- Product ratings are average across the board.** Averages range from 3.62 to 3.86 (overall 3.75), with Gloves, Sandals and Boots at the top.
- Faster shipping is linked to slightly higher spend.** Express customers spend 3.5% more per purchase than Standard customers.
- Young Adults contribute the most revenue (26.7%),** but all four age groups are close.

Evidence for finding 1: lowest and highest average purchase within each grouping

Average purchase	Gender	Category	Age group	Shipping type	Subscription
Lowest	\$59.54	\$57.17	\$59.07	\$58.46	\$59.49
Highest	\$60.25	\$60.26	\$60.45	\$60.73	\$59.87

7. Business Recommendations

Recommendation	Suggested actions	Based on
Rework subscription benefits	Add benefits that encourage larger or more frequent orders (such as free express shipping or member-only bundles), and target the 2,518 repeat buyers who have not subscribed.	Findings 4, 7
Reward loyal customers	Create a tiered loyalty programme for the 3,116 Loyal customers, using previous purchases to set the tiers.	Finding 3
Review the discount strategy	Check margins on Hat, Sneakers, Coat, Sweater and Pants, test smaller or targeted discounts, and offer premium deals to the 839 high-spending discount users.	Finding 5
Grow average order value	Because spend per order is flat, use bundles, cross-selling and free-shipping thresholds to increase basket size.	Finding 1
Strengthen core categories	Keep Clothing and Accessories well stocked and prominent, and grow Footwear and Outerwear through bundles with core items.	Finding 2
Improve product satisfaction	Collect feedback on the lowest-rated items (Shirt 3.62, Jeans 3.65, Blouse 3.68) and feature the top-rated Gloves, Sandals and Boots in campaigns.	Finding 6
Target campaigns by age	Tailor campaigns for Young Adults, the highest-revenue group, while keeping messaging broad because all age groups contribute similarly.	Finding 8

8. Limitations and Next Steps

- **Snapshot data:** each row is a customer's latest purchase with no dates, so trends over time and actual repeat-purchase timing cannot be measured.
- **Dataset quirk:** no female customer is a subscriber or used a discount. This is most likely a feature of this sample dataset, so no gender-based conclusions are drawn about discounts or subscriptions.
- **Even distributions:** season, payment method, shipping type and size are spread almost evenly, which suggests synthetic data and limits how much these factors can explain.
- **No sales channel column:** online vs offline behaviour could not be analysed directly.
- **Simple segment rules:** the New / Returning / Loyal thresholds (1, 2-10, more than 10) are rule-based; different cut-offs would change segment sizes.
- **Correlation, not causation:** differences such as higher spend with Express shipping show a relationship, not a cause.

Next steps

- **Add purchase dates** to track trends over time and measure how often customers really come back.
- **Test discount levels** with A/B experiments on the most discount-dependent products to find the best balance between sales and margin.

- **Predict subscription likelihood** with a simple classification model, so campaigns can target the repeat buyers most likely to subscribe.
- **Refine segmentation** using recency, frequency and monetary value (RFM) once transaction history is available.

9. Conclusion

The company's revenue is driven by the size of its customer base rather than by how much each customer spends, and most customers are already loyal repeat buyers. The biggest opportunities are to raise the value of each order, make the subscription programme more rewarding, and use discounts more selectively. Clothing and Accessories should remain the core focus, while bundles and cross-selling can grow the smaller categories.

Together, the Python preparation, SQL analysis and Power BI dashboard give the business a repeatable way to monitor these patterns and measure the impact of future marketing and product decisions.

Project files

File	Description
customer_shopping_behavior.csv	Original dataset
Customer_Shopping_Behaviour_Analysis.ipynb	Data preparation (Python)
customer_behaviour_sql_queries.sql	Business questions (SQL)
customer_behaviour_dashboard.pbix	Interactive dashboard (Power BI)
Customer_Shopping_Behaviour_Report.pdf	This report
Customer_Shopping_Behaviour_Presentation.pptx	Stakeholder presentation